SkillBridge AI: AI-Powered CV Skill Gap Analysis and Personalized Course Recommendations

Homework 1 | Project Selection & Exploratory Data Analysis — Updated Report

GitHub Repository: https://github.com/sudemkalip/SkillBridge_AI

1) Project Name

SkillBridge AI — Personalized Career Skill Gap and Learning Recommendation System

2) Overview

SkillBridge AI is a proposed AI-powered tool for university students and recent graduates. A user uploads a CV and selects a target occupation. The system extracts skills and relevant qualifications from the CV, retrieves the occupation's required skills and knowledge from the O*NET database, and compares the two profiles. It presents the results as Learn (no supporting evidence found), Improve (some evidence, but development may be needed), and Sufficient (evidence appears to meet the target level). For Learn skills, it recommends beginner courses; for Improve skills, it recommends intermediate or advanced courses. The final output is a personalized learning roadmap. The prototype is planned, not yet implemented.

The project supports SDG 4, Target 4.4 by helping users develop employment-related skills. It also contributes indirectly to SDG 8, Target 8.5 by supporting career readiness; it does not claim to guarantee employment.

3) Background

Students often have limited information about the specific skills required for their preferred occupations. A CV describes education, projects, experience, and certifications, but does not directly show how well those qualifications align with an occupation. At the same time, large online course catalogs make it difficult to choose an appropriate learning level. SkillBridge AI connects occupational skill requirements, individual CV evidence, and course information in one actionable process. Because CVs are incomplete representations of ability, the system will allow users to review and correct extracted skills and proficiency estimates.

4) Key Objectives / Business Objectives

a) Research Questions

- How can NLP identify occupationally relevant skills and qualifications in a user's CV?
- How can CV evidence be compared with O*NET occupational skill requirements?
- How can skill gaps be grouped into Learn, Improve, and Sufficient without overstating what a CV proves?
- How can skill gaps be matched to relevant courses at an appropriate difficulty level?
- How can we assess the relevance and reliability of the proposed recommendations?

b) Key Steps

- Select and examine occupational and course datasets; conduct exploratory data analysis.
- Clean and standardize occupation identifiers, skill labels, and course difficulty categories.
- Design a CV text extraction and skill identification component.
- Compare the CV profile with the selected occupation's skill profile and let users verify results.
- Recommend beginner courses for Learn and intermediate/advanced courses for Improve.
- Evaluate extraction, classification, and course relevance on manually reviewed examples; build a simple

end-to-end demonstration.

5) Methods and Workflow

a) Datasets

Dataset 1 — O*NET Occupational Database (official): O*NET 31.0 Occupation Data, Essential Skills, and Knowledge tables. Occupation codes connect job titles with required skills and knowledge. Importance measures describe occupational requirements, not an individual's demonstrated proficiency. The current release labels the relevant skills table "Essential Skills."

Dataset 2 — Coursera Course Dataset (third-party mirror): azrai99/coursera-course-dataset on Hugging Face. The downloaded 2024 Excel file includes course titles, skills, descriptions and levels. It is not an official Coursera data release. Course URLs, licenses, and current availability will need further verification.

User input: A target occupation and a CV (PDF or DOCX). User CVs are inputs to the eventual application, not part of the initial public EDA datasets.

b) Data Cleaning and Preprocessing — Performed in Homework 1

- Loaded four source tables in Python/pandas and checked row/column counts, total missing cells, and fully duplicated rows.
- Created separate cleaned copies of all four tables; applied duplicate-row removal (no duplicate rows were detected).
- Removed the non-informative "Unnamed: 0" index column from the Coursera table, reducing its columns from 14 to 13.
- Trimmed leading and trailing whitespace from Coursera title and Level values.
- Inspected missingness in key Coursera fields. Missing levels and descriptions were retained rather than guessed or silently removed.
- Further standardization of O*NET/Coursera skill terms and occupation-specific joins are planned for later stages; they have not yet been implemented.

c) Exploratory Data Analysis (Homework 1) — Completed Results EDA was conducted in Google Colab using pandas and matplotlib. The counts below are from the executed notebook and screenshots of its outputs.

Interpretation: The occupation table is complete in this initial check. The other tables contain blank cells, but missing-cell totals should not be interpreted as counts of unusable records. Some O*NET fields (e.g., confidence bounds and relevance flags) are supplementary and may be structurally unfilled for particular records.

Missingness in key Coursera fields

The missing-value plots also showed substantial blanks in Coursera Satisfaction Rate, Schedule and num_reviews. The O*NET Essential Skills missingness was concentrated in Not Relevant; O*NET Knowledge had missing entries in Not Relevant, Lower CI Bound, Upper CI Bound, Standard Error, Recommend Suppress, and N. These columns were inspected but not indiscriminately dropped.

O*NET: Average skill importance

Dataset overview (raw data)

Dataset	Rows	Columns	Missing cells	Duplicates

O*NET Occupations	1,016	3	0	0
O*NET Essential Skills	18,200	15	9,100	0
O*NET Knowledge	60,060	15	99,416	0
Coursera Courses	6,645	14	8,526	0

Missingness in key Coursera fields

Field	Missing values	Implication
title	0	Course names are populated.
Skills	0	Non-null strings can still represent empty lists (e.g., []).
Description	10	A small number of descriptions are unavailable.
Level	778 (11.7%)	Level-based recommendations require a missing-level policy.

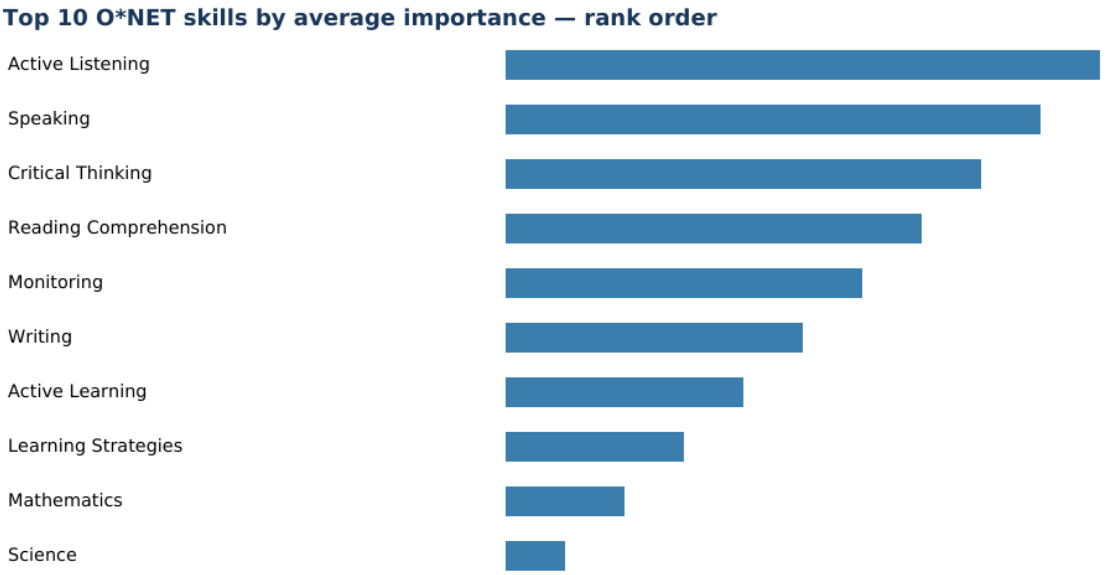
For Essential Skills, records with Scale ID = IM were selected and Data Value was averaged by Element Name across occupations. The highest-ranked skills were Active Listening, Speaking, Critical Thinking, Reading Comprehension, Monitoring, Writing, Active Learning, Learning Strategies, Mathematics and Science.

Figure 1. Rank-order visualization reconstructed from the notebook screenshot. Bar lengths indicate rank only; consult the notebook for exact mean scores. Cross-occupation averages do not replace target-occupation requirements.

Coursera: Course difficulty distribution The course-level distribution was dominated by Beginner courses, followed by Intermediate courses, with Advanced courses making up a much smaller group. The Level field was missing for 778 of 6,645 courses. This suggests that advanced-level recommendations may have fewer available candidates. Exact per-level counts are available from the notebook chart and should be exported before quoting numeric category totals.

Coursera: Skills covered by courses The most frequently tagged course skills were Data Analysis, Python Programming, Machine Learning, Data Visualization, Communication, Cloud Computing, Generative AI, Data Science, Project Management, and Artificial Intelligence (AI). The Skills field contains list-like strings; the EDA parsed these lists before counting tags.

O*NET average skill importance



Coursera course skills

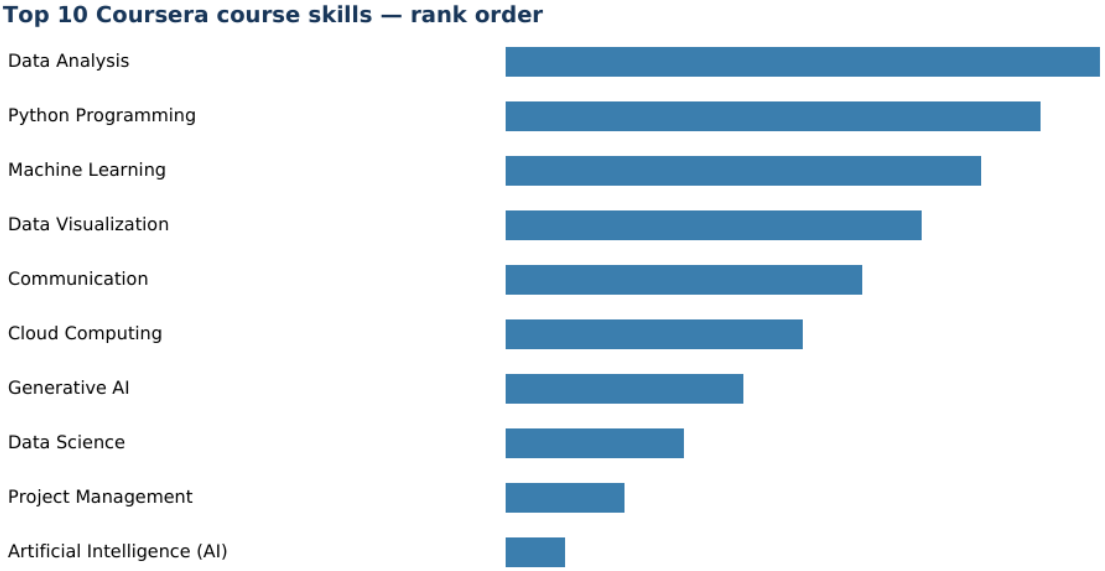


Figure 2. Rank-order visualization reconstructed from the notebook screenshot. Bar lengths indicate rank only; exact course counts should be taken from the notebook output. A course may contain multiple skill tags.

EDA Findings and Implications

- O*NET provides structured occupation-level skill requirements, while Coursera provides a candidate catalog of learning resources.
- The two datasets use different vocabularies: O*NET lists broad abilities such as Critical Thinking, while course tags include specific technologies such as Python Programming. Future matching will require semantic mapping, not exact string equality alone.
- The course catalog is uneven across levels and skill areas; missing course levels must be handled explicitly before personalized recommendations.

- Initial cleaning preserved records and source information. No predictive model, CV parsing, occupation-course matching, or end-to-end recommendation system has been trained or validated at the Homework 1 stage.

d) Proposed Modelling (Later Project Stages) Use PDF/DOCX text extraction followed by an NLP model to identify skills and qualifications. Use semantic text embeddings (for example, Sentence Transformers) to connect different skill expressions and match skills to course descriptions. A transparent decision layer will assign Learn / Improve / Sufficient labels based on CV evidence and user-confirmed proficiency; absence from a CV will be treated as "not demonstrated," not proof of no knowledge. Course ranking will consider semantic relevance and beginner/intermediate/advanced level. The model and thresholds will be chosen after initial tests.

e) Deliverables

- Homework 1: this concept note with completed initial EDA, selected public datasets, preprocessing steps, summary statistics, visualizations, and interpretations.
- Later stages: working CV-to-skill comparison prototype, course recommendations, evaluation on reviewed examples, and an ethical-risk assessment.

6) Expected Impact and Ethical Considerations

The tool aims to make learning choices more targeted and understandable. Evaluation will consider extraction precision/recall, agreement of classifications with reviewed cases, and the relevance and level of recommended courses. Personal CV data will be minimized, sensitive attributes excluded from recommendations, and unnecessary file retention avoided. Course availability and skill mappings will require verification. Recommendations will be presented as guidance, not definitive assessments of employability.

7) References and Dataset Sources

O*NET Resource Center. (n.d.). O*NET Database. https://www.onetcenter.org/database.html azrai99. (n.d.). Coursera Course Dataset. Hugging Face. https://huggingface.co/datasets/azrai99/coursera-course-dataset United Nations. (n.d.). Goal 4: Quality education. https://sdgs.un.org/goals/goal4 United Nations. (n.d.). Goal 8: Decent work and economic growth. https://sdgs.un.org/goals/goal8 Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. EMNLP-IJCNLP.

8) Code and Reproducibility

Google Colab notebook: SkillBridge_AI_Homework1_EDA.ipynb (to be uploaded to GitHub). The notebook includes dataset loading, missingness/duplicate checks, O*NET average-importance analysis, Coursera level and skill-tag analysis, figure generation, and initial preprocessing. GitHub repository URL: [ADD LINK AFTER UPLOAD].

Data provenance note: Dataset sizes and missingness counts are based on the user-provided Colab outputs. Figures 1 and 2 are qualitative rank reconstructions from screenshots, not exported original plots. Replace them with notebook PNG exports for a fully reproducible submission.